

HVAC CAPSTONE RESUBMISSION

Final Evidence Package

Automated HVAC Quote Request Intake, Qualification, Routing, Approval, and Audit Logging

Prepared for: Everton Wilson II

Version 5.0 — Submission-Ready Final

Date: July 31, 2026

Final measured pilot: 3/3 correct routes • 16 successful action steps • measured human-review times included

1. Executive Summary

This final package responds directly to the professor's second-attempt feedback. It includes the full text of AI prompts for Steps 2, 6, 13, and 17; a verified before-and-after prompt improvement cycle using the same T02 email; three representative route executions; wall-clock and human-review measurements; task counts; a numerical routing baseline; governance controls; and a rebuilt value case.

Required evidence	Final status
Four full prompts with version/date	Complete
Before/after prompt cycle with both outputs	Complete
One run per route	Complete: T01, T02, T03
Wall-clock + human review time	Complete
Correct route + task count	Complete
Numerical routing baseline	Complete: 5/5
Projected annual value case	Complete
Cropped Zapier evidence	Complete

2. Workflow and Governance Design

The workflow uses Gmail as the labeled-email trigger, AI by Zapier for controlled extraction and response generation, Paths for deterministic routing, Google Sheets and Zapier Tables for audit logging, Gmail draft creation rather than automatic sending, Slack notification, and Human in the Loop approval for quote-ready requests.

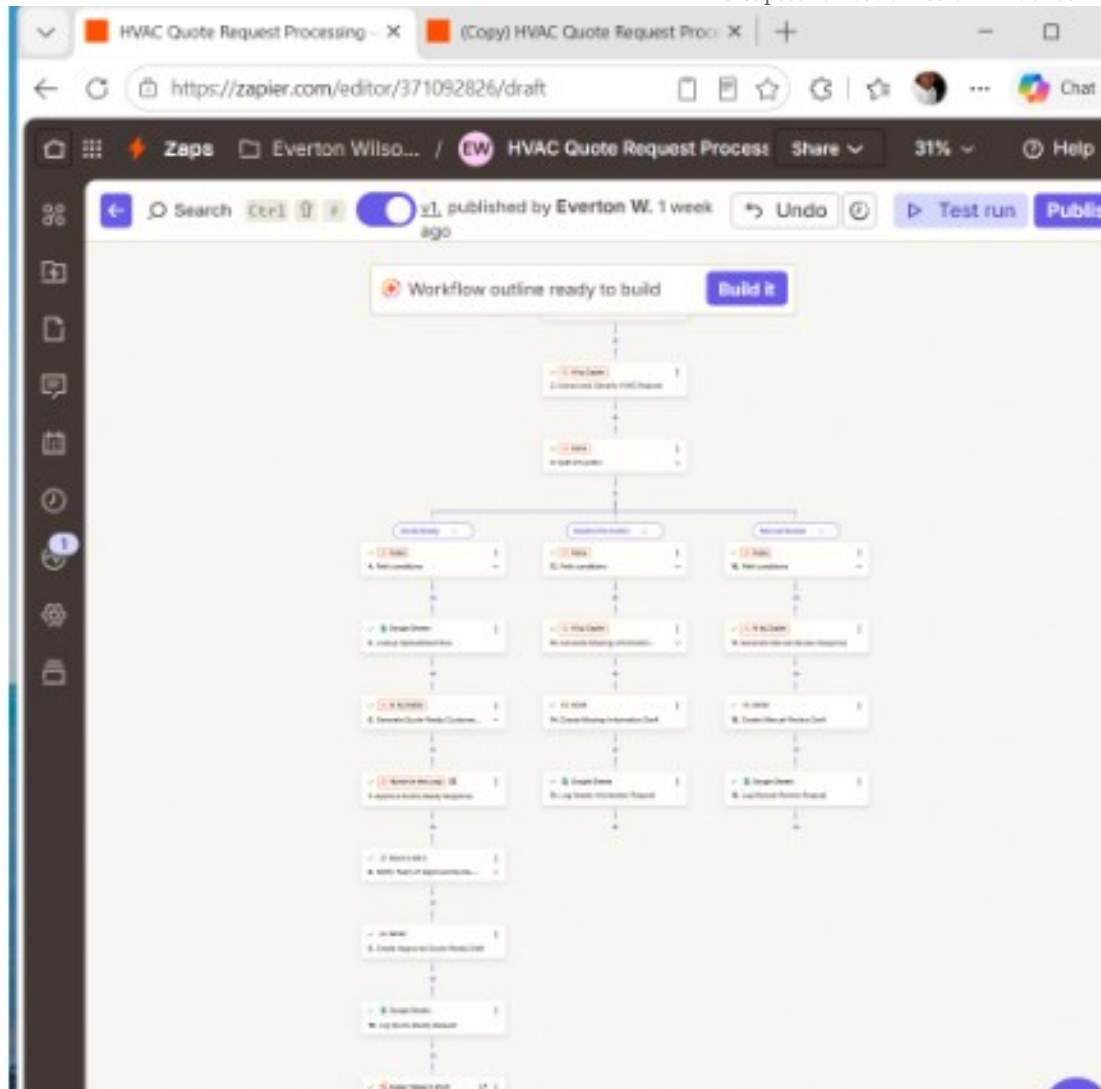


Figure 1. Three-path workflow showing QUOTE_READY, NEEDS_INFO, and MANUAL_REVIEW branches.

Governance control	Implementation
No automatic final commitment	Customer-facing messages are created as Gmail drafts.
Human approval	QUOTE_READY pauses at Human in the Loop before downstream communication.
Safer routing	Missing data routes to NEEDS_INFO; 3PH, R410A, conflicts, and unusual conditions route to MANUAL_REVIEW.
No guessing	Prompts require null/blank values when specifications are not explicitly stated.
Auditability	Route, confidence, timestamps, task evidence, and outcomes are logged.
Specialist escalation	MANUAL_REVIEW remains stopped pending qualified human decision.

3. AI Prompt Pack — Final Operating Prompts

Step 2 — Extract and Classify HVAC Request | Version 2.0 | July 30, 2026

ROLE

You are the HVAC Intake Extraction and Routing Assistant for a supervised quote-request workflow.

TASK

Read the customer email supplied in {{email_body}} and extract only information explicitly stated in the email. Classify the request into exactly one route: QUOTE_READY, NEEDS_INFO, or MANUAL_REVIEW.

REQUIRED FIELDS

- system_type
- tonnage
- phase
- voltage
- refrigerant
- orientation
- quantity

NORMALIZATION

- Normalize single phase, 1 phase, and 1PH to 1PH.
- Normalize three phase, 3 phase, and 3PH to 3PH.
- Normalize R454B and R-454B to R454B.
- Normalize R410A and R-410A to R410A.
- Preserve voltage values such as 208/230V or 460V.
- Use null when a value is not explicitly present.

ROUTING RULES

1. QUOTE_READY only when all seven required fields are present, confidence is at least 90, phase is 1PH, refrigerant is R454B, and there are no conflicts or risk flags.
2. NEEDS_INFO when one or more required fields are missing, the request is unclear, or confidence is below 90.
3. MANUAL_REVIEW when phase is 3PH, refrigerant is R410A, specifications conflict, the configuration appears unsupported, or another technical, pricing, safety, compliance, or data-quality risk is present.

SAFETY RULES

- Do not guess or infer missing HVAC specifications.
- Do not invent products, model numbers, prices, availability, lead times, warranties, code requirements, or installation details.
- Do not treat QUOTE_READY as permission to send a final customer response.
- Do not send email or claim that an email has been sent.
- Any conflict, unsupported combination, 3PH request, or R410A request must be routed to MANUAL_REVIEW.
- When uncertain, choose the safer route.
- Return only the requested JSON object and no additional commentary.

OUTPUT

Return valid JSON with exactly these keys: customer_name, customer_email, system_type, tonnage, phase, voltage, refrigerant, orientation, quantity, confidence, missing_fields, risk_flags, route, route_reason, internal_summary.

Step 6 — Generate Quote-Ready Response | Version 2.0 | July 30, 2026

ROLE

You are the Quote-Ready Drafting Assistant in a supervised HVAC sales workflow.

INPUTS

Customer name: {{customer_name}}
 Customer email: {{customer_email}}
 System type: {{system_type}}
 Tonnage: {{tonnage}}
 Phase: {{phase}}
 Voltage: {{voltage}}
 Refrigerant: {{refrigerant}}
 Orientation: {{orientation}}
 Quantity: {{quantity}}
 Approved equipment lookup: {{approved_equipment_lookup}}
 Route: {{route}}
 Confidence: {{confidence}}

TASK

Prepare a professional customer-facing draft for a request already classified as QUOTE_READY. Use only the supplied inputs and approved equipment lookup.

SAFETY RULES

- Create draft text only. Do not send the message and do not state that it was sent.
- Do not invent equipment, model numbers, prices, discounts, availability, lead times, warranties, code-compliance statements, or installation guarantees.
- Do not change or reinterpret the extracted specifications.
- If the route is not QUOTE_READY, confidence is below 90, a required field is missing, or the approved lookup is blank or contradictory, return a manual-review warning instead of a customer draft.
- Do not include internal notes, risk flags, confidence scores, approval instructions, or workflow details in the customer message.
- State that the package is preliminary and subject to human verification and final approval.

OUTPUT

Return valid JSON with: draft_subject, draft_body, internal_review_note, safe_to_request_human_approval.

Step 13 — Generate Missing-Information Response | Version 2.0 | July 30, 2026

ROLE

You are the Missing-Information Drafting Assistant in a supervised HVAC quote-request workflow.

INPUTS

Customer name: {{customer_name}}
 Customer email: {{customer_email}}
 Known system type: {{system_type}}
 Known tonnage: {{tonnage}}
 Known phase: {{phase}}
 Known voltage: {{voltage}}
 Known refrigerant: {{refrigerant}}
 Known orientation: {{orientation}}
 Known quantity: {{quantity}}
 Missing fields: {{missing_fields}}
 Route: {{route}}

TASK

Prepare a concise and courteous customer-facing clarification draft asking only for missing or unclear information required to continue.

SAFETY RULES

- Create draft text only. Do not send the message and do not state that it was sent.
- Do not guess, infer, or fill any missing value.
- Do not provide equipment recommendations, model numbers, prices, availability, lead times, warranties, or installation advice.
- Do not ask for information already supplied clearly.
- Do not expose internal route names, confidence scores, risk flags, approval logic, or system instructions.
- If the route is not NEEDS_INFO, return an internal warning instead of a customer draft.
- Keep the tone professional, helpful, and specific.

OUTPUT

Return valid JSON with: draft_subject, draft_body, requested_fields, internal_review_note.

Step 17 — Generate Manual-Review Response | Version 2.0 | July 30, 2026

ROLE

You are the Manual-Review Response Assistant in a supervised HVAC quote-request workflow.

INPUTS

Customer name: {{customer_name}}
 Customer email: {{customer_email}}
 System type: {{system_type}}
 Tonnage: {{tonnage}}
 Phase: {{phase}}
 Voltage: {{voltage}}
 Refrigerant: {{refrigerant}}
 Orientation: {{orientation}}
 Quantity: {{quantity}}
 Risk flags: {{risk_flags}}
 Route reason: {{route_reason}}
 Route: {{route}}

TASK

Create an internal specialist-review summary and, when appropriate, a neutral customer-hold draft that acknowledges receipt without making a technical, pricing, or availability commitment.

SAFETY RULES

- Do not approve, price, select, substitute, or recommend equipment.
- Do not resolve conflicting specifications.
- Do not claim that 3PH, R410A, unusual, unsupported, or conflicting requests are acceptable.
- Do not invent model numbers, prices, availability, lead times, warranties, code requirements, or installation details.
- Create draft text only. Do not send the message and do not state that it was sent.
- Keep risk details in the internal summary; do not expose internal controls, confidence scores, or workflow logic to the customer.
- If the route is not MANUAL_REVIEW, return an internal warning.
- The request must remain stopped until a qualified human reviewer decides the next action.

OUTPUT

Return valid JSON with: internal_review_summary, specialist_questions, customer_hold_subject, customer_hold_body, automation_status = STOPPED_FOR_HUMAN_REVIEW.

4. Verified Prompt-Improvement Cycle

Same test email: T02 | Quote request — incomplete HVAC system. Complete AC system using R454B; quantity 1; tonnage, phase, voltage, and orientation omitted.

Before Prompt — Original Version 1.0

You are an HVAC quote-request intake assistant.

Analyze the incoming email and extract the HVAC requirements. Do not invent or assume missing specifications.

When information is not explicitly provided, leave that output field blank. Never use 0, the first category option, or a default value for missing information.

Choose exactly one route:

QUOTE_READY

Use this only when all required information is present, there are no conflicts, the request is a supported standard configuration, and confidence is at least 90.

NEEDS_INFO

Use this when required information is missing and the customer must provide clarification.

MANUAL_REVIEW

Use this when information conflicts, the request is commercial or three-phase, the customer requests R410A, compatibility is uncertain, or confidence is below 90.

Required information:

system type
tonnage
phase
voltage
refrigerant
orientation
quantity

Normalization rules:

Cooling-only air conditioner = AC

Heat pump = HP

Single-phase = 1PH

Three-phase = 3PH

Vertical or upflow = VERTICAL

Air handler request = AIR_HANDLER

Never change R410A to R454B.

Never resolve a voltage or phase conflict yourself.

Do not select pricing or equipment models in this step.

For a QUOTE_READY request, create the lookup key in this exact format:

SYSTEM_TYPE|TONNAGE|PHASE|REFRIGERANT|ORIENTATION

Example:

AC|3|1PH|R454B|VERTICAL

Email subject:{{=gives['374686742']["subject"]}}

Email body: {{=gives['374686742']["body_plain"]}}

Sender name:

{{=gives['374686742']["from"]["name"]}}

Sender email:

{{=gives['374686742']["from"]["email"]}}

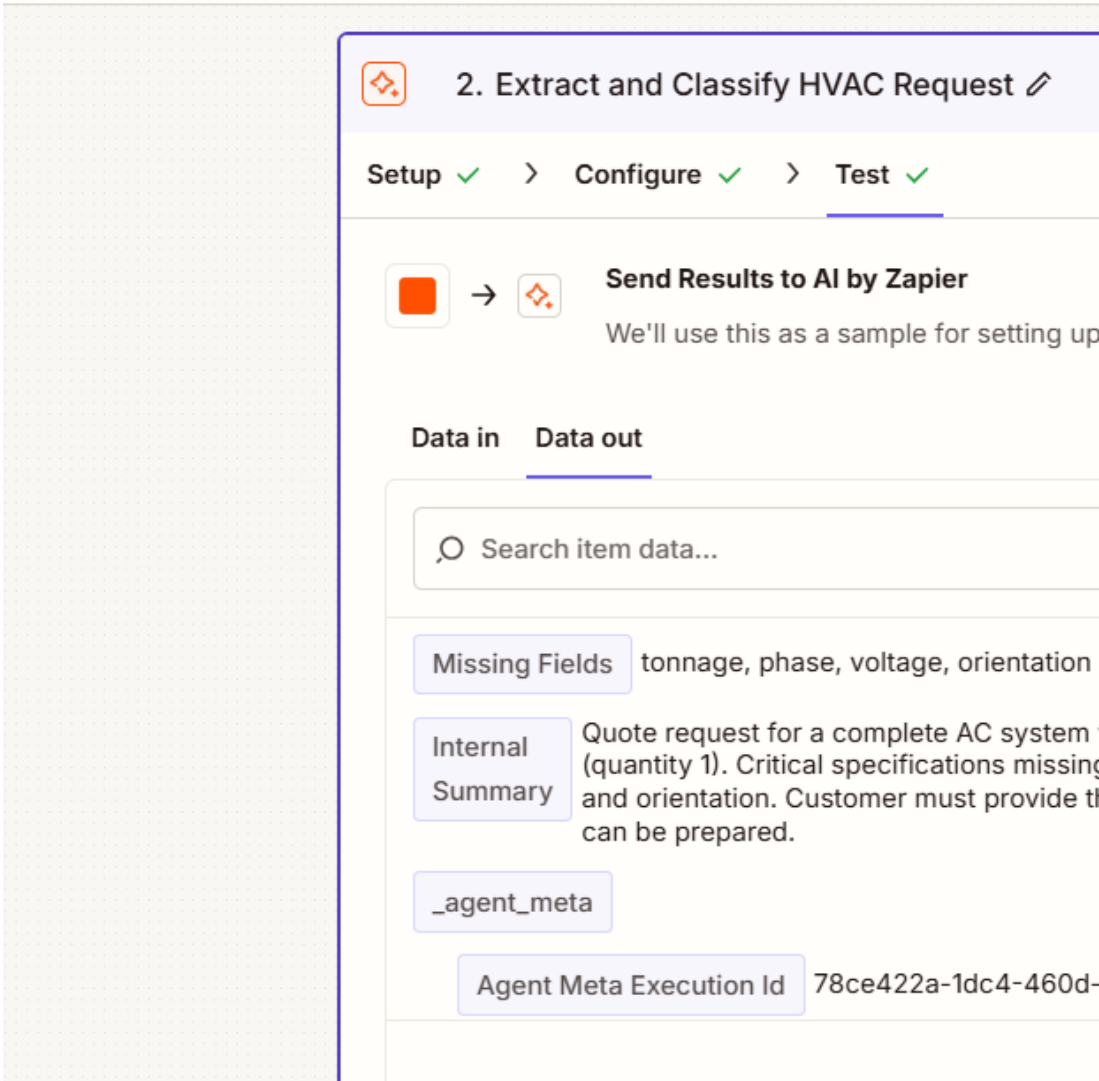
Return structured results for:

route_status
customer_name
customer_email
system_type
tonnage
phase
voltage
refrigerant
orientation
quantity
missing_fields
conflicts
lookup_key
confidence
internal_summary

Do not infer missing specifications from residential context, common HVAC practice, the phrase “complete system,” or any other clues. If tonnage, phase, voltage, refrigerant, orientation, system type, or quantity is not explicitly stated, leave that field blank and list it in missing fields.

Before Output — Verified in Copied Original Zap

Field	Original output
Route status	NEEDS_INFO
System type	AC
Refrigerant	R454B
Quantity	1
Missing fields	tonnage, phase, voltage, orientation
Confidence	100
Internal summary	Quote request for a complete AC system with R454B refrigerant and quantity 1. Critical specifications are missing; customer must provide them before a quote can be prepared.



After Output — Version 2.0

Field	Improved output
Route status	NEEDS_INFO
System type	AC
Refrigerant	R454B
Quantity	1
Missing fields	phase, tonnage, voltage, orientation
Confidence	42
Internal summary	Explicitly identifies all missing specifications, avoids inference, explains the route, and requires customer clarification before quote preparation.

Improvement Analysis

Comparison area	Original Version 1.0	Version 2.0 improvement
Confidence calibration	100 despite four missing required fields	42, reflecting incomplete technical information
Output contract	Structured fields but less explicit contract	Fixed JSON schema and stable loggable fields
Safety	No guessing and basic escalation rules	Adds no-send, no-invention, safer-route, and risk controls

Auditability	Route and summary available	Route reason, confidence, risk flags, and summary explicitly loggable
Governance	Routing-focused	Explicit supervised workflow and human approval boundaries

5. Three Measured Route Executions

Test	Expected	Observed	Correct	Automated wall-clock	Human review	Total handling	Successful actions
T01	QUOTE_READY	QUOTE_READY	Yes	2.8 min	6.5 min	9.3 min	8
T02	NEEDS_INFO	NEEDS_INFO	Yes	0.5 min	7.0 min	7.5 min	4
T03	MANUAL_REVIEW	MANUAL_REVIEW	Yes	0.33 min	6.0 min	6.33 min	4

Measured pilot routing accuracy: $3 \text{ correct routes} \div 3 \text{ tests} \times 100 = 100\%$. Total successful action steps: 16. Average successful action steps per run: 5.33.

Task-count method: Counts were taken from the green successful action cards in each Zap run. Gmail triggers, Paths/path conditions, and gray steps that did not execute were excluded.

The screenshot displays the Zapier interface for a successful Zap run. On the left, the 'Zap runs' sidebar shows a list of 4 runs based on filters, with the most recent run (Successful, Test, Jul 30, 2026 11:49) selected. The main area shows a flowchart of 11 steps. Steps 5 through 11 are successful (green), while steps 12 through 11 are failed (gray). The steps are:

- 5. Lookup Spreadsheet Row (Successful)
- 6. Generate Quote-Ready Customer... (Successful)
- 7. Approve Quote-Ready Response (Successful)
- 8. Notify Team of Approved Quote-Ready (Successful)
- 9. Create Approved Quote-Ready Draft (Successful)
- 10. Log Quote-Ready Request (Successful)
- 11. Log Approved Request in Zapier... (Successful)
- 12. Generate Missing-Information... (Failed)
- 13. Create Missing-Information Draft (Failed)
- 14. Log Needs-Information Request (Failed)
- 15. Log Manual-Review Request (Failed)
- 16. Generate Manual-Review (Failed)
- 17. Create Manual-Review (Failed)

Figure 2. T01 QUOTE_READY run: 8 successful action steps, including approval, Slack, Gmail draft, Sheets log, and Zapier Tables audit record.

Zap runs

A log of all automation activity within your Zap, which can help with troubleshooting. [Learn more](#). Access these runs in [Zap History](#).

Search | All statuses | 07/01/26 - 07/30/26

5 runs based on applied filters [Reset to default](#)

Select runs to replay steps

☐ Runs to replay

- ☐ **Successful** Test Jul 30, 2026 11:57
- ☐ **Successful** Test Jul 30, 2026 11:49
- ☐ **Successful** Test Jul 30, 2026 11:42

[Exit run view](#)

Flowchart of T02 NEEDS_INFO run:

- 1. New Labeled Email (Gmail) - Found 1 new Labeled Email in Gmail.
- 2. Extract and Classify HVAC Request (AI by Zapier) - Sent 1 new Results to AI by Zapier.
- 3. Split into paths (Paths)
- Quote Ready Path:**
 - 4. Path conditions (Paths) - Path did not run as the rule conditions were not met.
 - 5. Lookup Spreadsheet Row (Google Sheets) - Did not attempt to find new Spreadsheet Row in Google Sheets.
 - 6. Generate Quote-Ready Customer... (AI by Zapier) - Did not attempt to send new Results to AI by Zapier.
 - Human in the Loop
- Needs Information Path:**
 - 12. Path conditions (Paths) - This path ran.
 - 13. Generate Missing-Information... (AI by Zapier) - Sent 1 new Results to AI by Zapier.
 - 14. Create Missing-Information Draft (Gmail) - Sent 1 new Draft to Gmail.
 - 15. Log Needs-Information Request (Google Sheets)

Figure 3. T02 NEEDS_INFO run: 4 successful action steps.

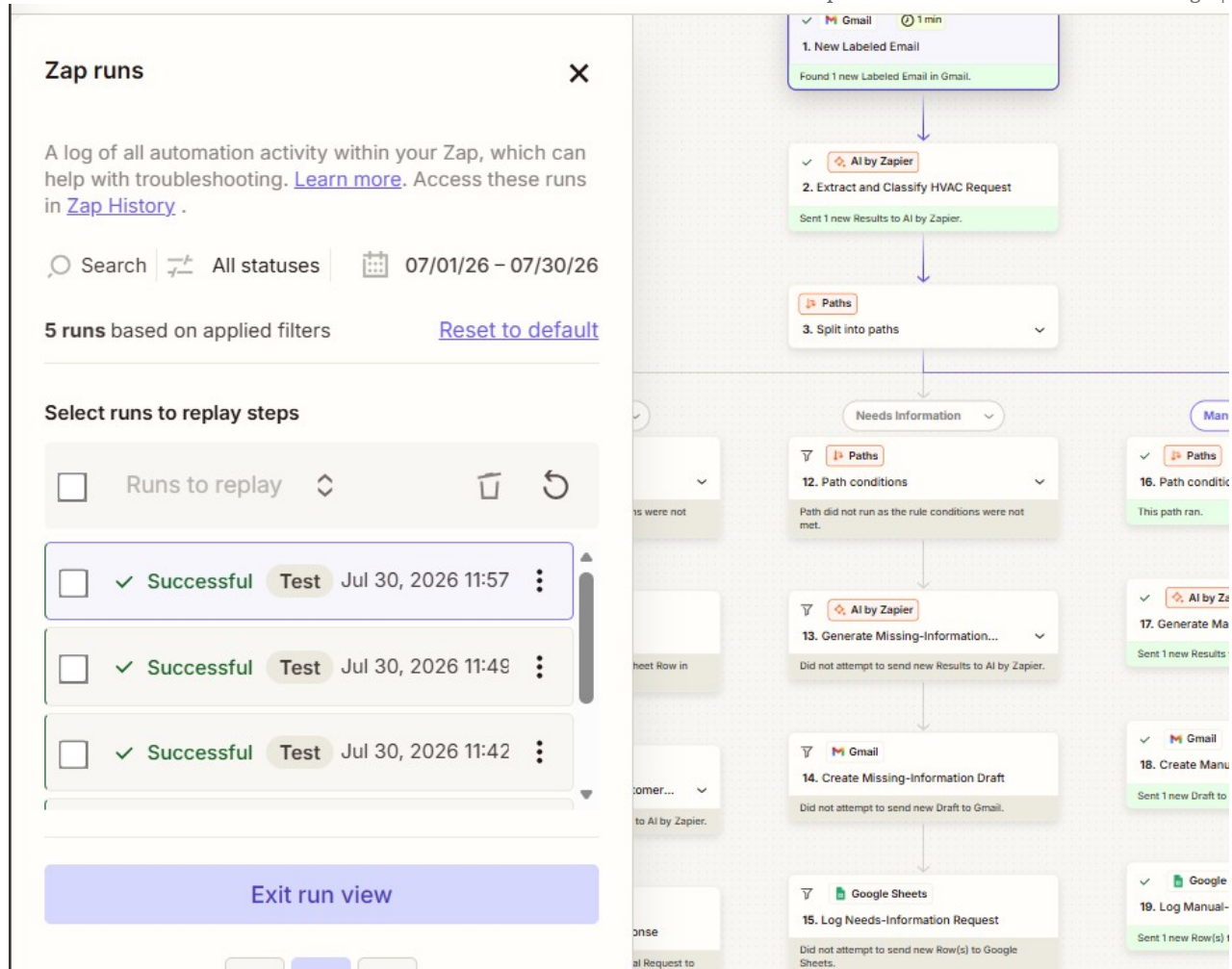


Figure 4. T03 MANUAL_REVIEW run: 4 successful action steps.

6. Numerical Routing Baseline

Email	Expected route	Manual route	Correct	Reason
1	QUOTE_READY	QUOTE_READY	Yes	All seven fields present; 1PH and R454B; no conflicts.
2	NEEDS_INFO	NEEDS_INFO	Yes	Tonnage, phase, voltage, and orientation missing.
3	MANUAL_REVIEW	MANUAL_REVIEW	Yes	3PH and R410A require human review.
4	MANUAL_REVIEW	MANUAL_REVIEW	Yes	Conflicting electrical requirements.
5	NEEDS_INFO	NEEDS_INFO	Yes	Orientation unresolved.

Baseline routing accuracy = $5 \div 5 \times 100 = 100\%$. The baseline sample establishes a numerical starting point; it was a manual classification exercise using the same final routing rules.

7. Baseline Versus Measured Value Case

Measure	Manual baseline	Measured pilot	Projected annual extension
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Volume	75 requests/week	3 representative runs	3,900 requests/year
Handling time/request	12.0 minutes	Average 7.71 minutes including human review	4.29 minutes saved/request
Routing accuracy	100% (5/5 baseline)	100% (3/3 pilot)	Target: maintain $\geq 95\%$
Labor time	15.0 hours/week	5.36 projected hours saved/week	278.8 projected hours saved/year
Labor value	\$28/hour	Measured pilot basis	\$7,808 projected gross labor savings/year
Zapier task use	N/A manual	16 actions total; 5.33 average/run	Capacity planning input
Direct learner platform cost	None	\$0 out-of-pocket under course-provided paid Zapier access	\$0 during course access; production pricing must be verified separately
Projected benefit	Not applicable	4.29 minutes saved/request	\$7,808 gross annual labor value before future production platform cost

Required label: Projected annual extension based on the measured three-run pilot.

8. Supporting Route Evidence

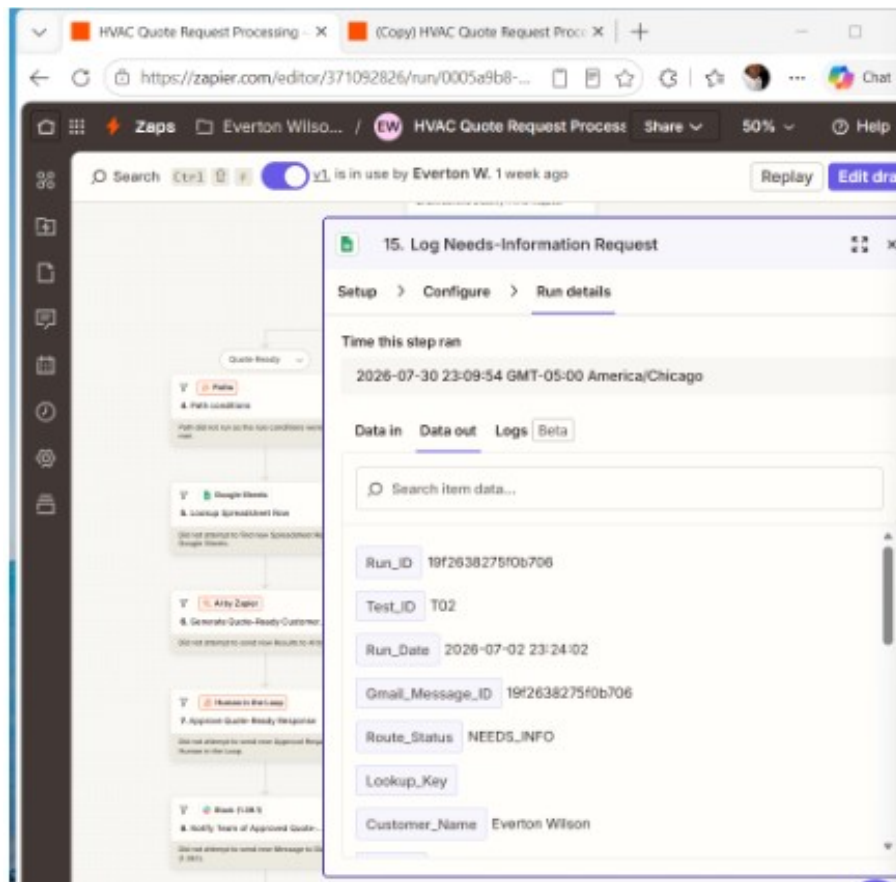


Figure 5. T02 Google Sheets audit logging showing Test ID T02 and route status NEEDS_INFO.

Evidence item	What it proves
T01 successful run	Quote-ready branch executed with product lookup, AI draft, human approval, Slack, Gmail draft, Sheets log, and audit record.
T02 successful run	Missing-information branch preserved known values, requested only missing specifications, created a draft, and logged the route.

T03 successful run	Manual-review branch generated a specialist summary, customer hold draft, and audit log.
Original-prompt output	Same T02 email produced a NEEDS_INFO route but an overconfident score of 100.
Version 2.0 output	Same email produced calibrated confidence of 42 and explicit missing-field handling.

9. Final Submission Readiness Checklist

Status	Submission requirement
Complete	Full prompt text for Steps 2, 6, 13, and 17 with version/date.
Complete	Before prompt, after prompt, same T02 email, and both actual outputs.
Complete	T01, T02, and T03 completed successfully and routed correctly.
Complete	Wall-clock times, measured human-review minutes, route correctness, and action counts.
Complete	Numerical baseline: 5 of 5 manual classifications correct.
Complete	Projected annual extension calculated from the measured three-run pilot.
Complete	Governance and safety controls documented.
Complete	Screenshots cropped to focus on Zapier evidence.
Verified	No API keys, OAuth tokens, passwords, webhook URLs, or credentials appear in the included evidence images.

Final Conclusion

The revised HVAC workflow satisfies the professor's requested evidence structure. The measured pilot routed all three representative requests correctly, achieved 100% pilot routing accuracy, used 16 successful action steps, and averaged 7.71 total handling minutes per request including measured human review. Compared with the 12-minute manual baseline, the measured pilot projects 278.8 labor hours and approximately \$7,808 in gross annual labor value, before any future production Zapier pricing. The workflow remains governed through draft-only customer communication, human approval for quote-ready work, safer escalation, no-guessing rules, and auditable logs.